

EMCH 792: Optimal State Estimation
Homework 3
Computer Portion Solutions

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Problem B1: Exercise 2.15 — Uniform Random Variables (20 pts)

Generate independent random variables uniformly distributed on $(0, 1)$ for $N = 50, 500, 5000$. Plot 10-bin histograms and report sample mean/standard deviation. Compare against theoretical values:

$$\mu = 0.5, \quad \sigma = \frac{1}{\sqrt{12}} \approx 0.288675$$

MATLAB Code

```

1 rng(792);
2 N_values = [50, 500, 5000];
3 theory_mean = 0.5;
4 theory_std = 1/sqrt(12);
5
6 for i = 1:numel(N_values)
7     N = N_values(i);
8     u = rand(N, 1);
9     sample_mean(i) = mean(u);
10    sample_std(i) = std(u);
11 end

```

MATLAB Output

```

Theoretical mean = 0.500000, theoretical std = 0.288675
N= 50 -> sample mean = 0.466955, sample std = 0.248034
N= 500 -> sample mean = 0.497011, sample std = 0.286045
N=5000 -> sample mean = 0.508207, sample std = 0.289937

```

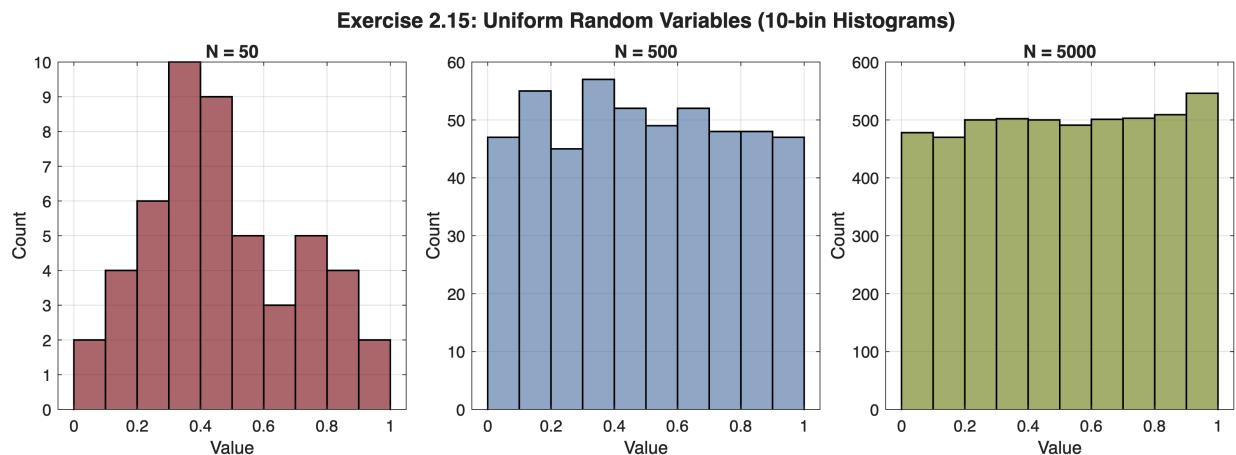


Figure 1: Exercise 2.15 histograms (10 bins) for $N = 50, 500, 5000$.

Results

As N increases, the sample statistics converge toward the theoretical values and the histogram shape becomes smoother and more representative of the uniform distribution.

Problem B2: Noise Covariance Matrix Calculation (20 pts)

Given

$$\dot{x}(t) = Ax(t) + B_u u(t) + B_w w(t),$$

$$A = \begin{bmatrix} -1 & 1 \\ -2 & -2 \end{bmatrix}, \quad B_u = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad B_w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad S_w = 0.11.$$

with $\Delta t_1 = 0.5$ and $\Delta t_2 = 0.01$.

Define the continuous covariance intensity:

$$Q_c = B_w S_w B_w^T = \begin{bmatrix} 0.11 & 0.22 \\ 0.22 & 0.44 \end{bmatrix}.$$

B2.1: Exact and Approximate $\Sigma_{\tilde{w}}$

Using Van Loan's block matrix:

$$Z = \begin{bmatrix} -A & Q_c \\ 0 & A^T \end{bmatrix} \Delta t,$$

$$\exp(Z) = \begin{bmatrix} E_{11} & E_{12} \\ 0 & E_{22} \end{bmatrix}, \quad A_d = E_{11}^{-1}, \quad \Sigma_{\tilde{w}}^{\text{exact}} = A_d E_{12},$$

and first-order approximation $\Sigma_{\tilde{w}}^{\text{approx}} = Q_c \Delta t$.

MATLAB Output

```
dt = 0.500
Sigma_w_exact =
    0.0590    0.0507
    0.0507    0.0591
Sigma_w_approx =
    0.0550    0.1100
    0.1100    0.2200
||difference||_F = 0.1814999427

dt = 0.010
Sigma_w_exact =
    0.0011    0.0022
    0.0022    0.0043
Sigma_w_approx =
    0.0011    0.0022
    0.0022    0.0044
||difference||_F = 0.0001341646
```

Results

The approximation error is large at $\Delta t = 0.5$ but very small at $\Delta t = 0.01$, confirming that $Q_c \Delta t$ is only accurate for sufficiently small sampling periods.

B2.2: Discrete A_d and $c2d()$ Verification

MATLAB Output

```
dt = 0.500
Ad (Z matrix) =
    0.4824    0.2193
   -0.4387    0.2631
Ad (c2d) =
    0.4824    0.2193
   -0.4387    0.2631
||Ad_Z - Ad_c2d||_F = 5.5511151231e-17

dt = 0.010
Ad (Z matrix) =
    0.9900    0.0099
   -0.0197    0.9801
Ad (c2d) =
    0.9900    0.0099
   -0.0197    0.9801
||Ad_Z - Ad_c2d||_F = 1.1102230246e-16
```

Results

A_d from the Z -matrix method matches MATLAB's $c2d()$ to machine precision for both sample times.

B2.3: Continuous/Discrete Steady-State Covariances

Continuous Lyapunov equation:

$$A\Sigma_{x,ss}^{(c)} + \Sigma_{x,ss}^{(c)}A^T + Q_c = 0.$$

Discrete Lyapunov equation:

$$\Sigma_{x,ss}^{(d)} = A_d\Sigma_{x,ss}^{(d)}A_d^T + \Sigma_{\tilde{w}}^{\text{exact}}.$$

MATLAB Output

```
Sigma_x_ss_cont =  
    0.0917    0.0367  
    0.0367    0.0733  
  
dt = 0.500  
Sigma_x_ss_disc =  
    0.0917    0.0367  
    0.0367    0.0733  
||disc - cont||_F = 3.6717175287e-17  
  
dt = 0.010  
Sigma_x_ss_disc =  
    0.0917    0.0367  
    0.0367    0.0733  
||disc - cont||_F = 3.3001717888e-16
```

Results

When the exact discrete process covariance is used, continuous and discrete steady-state covariance predictions agree (numerical differences only at machine precision).